

B. Tech Degree V Semester (Supplementary) Examination July 2009

ME 504 THERMAL ENGINEERING (2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART - A

(Answer ALL questions)

- I. (a) Derive an expression to show that the thermal efficiency of the air standard Otto cycle is a function of the compression ratio only.
- (b) Describe briefly the heat balance sheet chart for an IC engine.
- (c) What are the different types of cooling systems used in IC engine?
- (d) Explain Pre-ignition in spark ignition engines.
- (e) Explain various types of Steam nozzles.
- (f) Describe the working principle of impulse turbine and reaction turbine.
- (g) Suggest a thermodynamic cycle for a gas turbine. Derive an equation for its cycle efficiency.
- (h) Discuss the various method to improve the performance of a gas turbine.

PART - B

(4 x 15 = 60)

- II. (a) Draw and explain the valve timing diagram for a typical four stroke IC engine. (7)
- (b) A four stroke engine working on diesel cycle has a piston diameter of 25 cm, a stroke of 40cm and a clearance volume of 1200 CC. The fuel injection takes place for 5% of stroke. If the induction pressure corresponds to 1 bar and engine turns 5 rev/s, find the air standard efficiency and power developed. (8)

OR

- III. (a) What is Morse Test? Explain its step by step procedure. (6)
- (b) In a constant speed compression ignition engine operating on four-stroke cycle and fitted with band brake, the following observations were taken.

Brake wheel diameter	=	60 cm
Band thickness	=	5mm
Speed	=	450 rpm
Load on band	=	210 N
Spring balance reading	=	30 N
Area of indicator diagram	=	4.15cm ²
Length of indicator diagram	=	6.25cm
Spring No.	=	11 ie. 11 bar/cm
Bore	=	10 cm
Stroke	=	15 cm
Specific fuel consumption	=	0.3kg/KW-hr
Heating value of fuel	=	41800 kJ/Kg

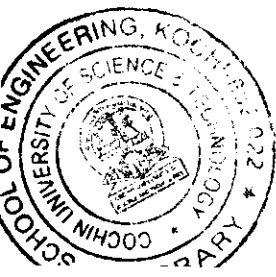
Determine the brake power, mechanical efficiency, the indicated thermal efficiency and the brake thermal efficiency. (9)

- IV. (a) Explain the working principle of Wankel engine. (6)
- (b) What are the different lubrication systems used in IC engines? (6)
- (c) Describe the merits of super charging in IC engines. (3)

OR

- V. Explain the combustion phenomenon in SI engines. (15)

(Turn Over)



- VI. 5 Kg of steam per minute pass through a convergent – divergent nozzle. The pressure and temperature of steam supplied to the nozzle box is 10 bar and 200°C respectively. The discharge pressure is 0.1 bar. The expansion is super – saturated upto throat and in thermal equilibrium afterwards. Calculate –

- (i) the area of the nozzle at exit
- (ii) the maximum degree of super saturation
- (iii) the degree of under cooling at the throat

Assume $pv^{1.3} = \text{constant}$ for super – saturation flow.

$$\text{Also } \frac{P}{T^{\frac{1.3}{0.3}}} = \text{constant for super – saturated flow.} \quad (15)$$

OR

- VII. The outlet angle of the blade of a Parson's turbine is 20° and the axial velocity of flow of steam is 0.5 times the mean blade velocity. If the diameter of the ring is 1.25 m and the rotational speed is 3000 rpm, determine the inlet angles of the blades. Determine also the power developed if dry saturated steam at 5 bar passes through the blades where height may be assumed as 6 cm. Neglect the effect of the blade thickness. (15)

- VIII. In a gas turbine installation, the air enters the compressor at 1 bar and leaves 4 bar and 500K. Fuel of calorific value of 44,000 KJ/Kg is burnt in the combustion chamber, 50 Kg of air being supplied for 1 Kg of fuel. The material of the turbine blades allows. For gases at a maximum temperature of 1000° K. Determine the area of the nozzles at throat and exit for mass flow of 8 Kg/s.

Assume $C_p = 1 \text{ kJ/Kg}^\circ\text{C}$ and $\gamma = 1.4$ for air

$C_p = 1.05 \text{ kJ/KgK}$ and $\gamma = 1.3$ for product of combustion. (15)

OR

- IX. The following data are given for a centrifugal compressor and it is required to find the impeller tip diameter.

N	=	15,000 rpm,
G	=	22.5 Kg/s.
Ambient temperature	=	21°C
Atmospheric pressure	=	1 bar
Velocity of flow at inlet	=	Exit flow from diffuser = 140 m/s
Pressure ratio	=	4
Adiabatic efficiency	=	0.76
Double sided impeller		
slip factor	=	0.9
Work input factor	=	1.04
Assume C_p as 1.005 kJ/Kg K.		

(15)
